

COM435 - Software Product and Process Management 2024-25

Lab Practical

Week 2

Task 1 – Finalise researching existing solutions

In week 1 each team should have identified several existing solutions for a “live public transport system”. Depending on the extent of that research you may need to continue identifying existing solutions and completing that research. Do not limit the research to Northern Ireland, instead try to find solutions further afield that might be more advanced in their offerings.

Using the identified solutions, which can be allocated to individual teams members or pairs, a list of features or functionalities should be listed for each solution.

Teams should then combine their features into a single list of features noting which features are common and perhaps how common they are (you can achieve this by noting how many times a particular feature is found across multiple solutions). You may also need to describe features that are otherwise vague to help the team understand what you have found.

Task 2

Having produced a list of features which represents a number of solutions, it's time to begin the evaluation stage. Perform a SWOT analysis on the existing solutions and list of features you have discovered to highlight their strengths and weaknesses and to identify any potential opportunities. It's less important to look at threats as at this stage you don't have a well-established product as the basis for which to test any threats.

By the end of week 2 you should, both individually and as a team, have a better understanding of the problem domain and what current solutions are offering.

In week 3, with this knowledge in mind you will start to look at the problem area from a particular stakeholder perspective using a number of value proposition canvases. This should lead to some additional new features being identified and also confirm a number of features that belong to existing solutions.

Begin by identifying a number of customer cohorts that will either directly use the web app or will benefit indirectly from its use by others. List these users and discuss their ranking in terms of which users will be the early adopters or have the most to gain from using the app. Starting with the number one ranked user take each user cohort in turn to explore what tasks are important to them in the context of a primary school app. You will need to empathise from the user's viewpoint which is not easy so try to structure your thinking logically. For example you might walk through a user's day from getting up in the morning, going through the school's schedule to sending the pupils home each day. Then start to think about how this plays out across the week and school term. Also think about what activities take place in school, when they occur, how important they are, who is involved and how could the web app help. Identify daily activities, weekly activities and those that occur less frequently or are tied to particular events such as half-term, Halloween, Christmas, fun raising events etc..

When initially exploring these ideas try not to be judgemental and allow ideas to flow. You will get a chance later to review, refine and evaluate what's going to be in versus out.

If you feel you need more time to complete the research activity from the previous lab please take more time to complete this before moving into this area. Alternatively, you can always switch back and forth between research and generating new ideas. Remember this is your team and each team will work at a slightly different pace.